

Online Fair Allocation of Indivisible Goods

Recent Progress and Open Questions

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Fall 2026

Outline

- 1 Online Models
- 2 Online Mechanisms & Problems
 - LIKE-Type Online Mechanisms
 - Vanishing Envy
 - EF1 with Reallocation
 - Online MMS
 - Restricted Valuations
 - Bandit Learning for Online Fair Allocation
 - Stochastic and Learning Models
 - Nash Welfare and the PACE Algorithm
 - Online Fair Division with Subsidies
- 3 Related Directions
- 4 Open Problems

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Canonical Online Goods Model

At each time $t = 1, 2, \dots, T$:

- 1 one indivisible good g_t arrives;
- 2 values $(v_1(g_t), v_2(g_t), \dots, v_n(g_t))$ are revealed, estimated, or partially observed;
- 3 the algorithm assigns g_t to one agent;
- 4 in the irrevocable model, this decision cannot be changed.

Final allocation:

$$A_i^T = \{g_t : g_t \text{ was assigned to } i\}.$$

Four Axes of Online Fair Allocation

1. Arrival model

adversarial / stochastic / predicted

2. Commitment model

irrevocable / batched / reallocation

3. Information model

full values / binary likes / bandit feedback

4. Fairness horizon

final-time / prefix-wise / asymptotic

An online fair-allocation model is specified by choosing one alternative along each axis.

Fairness Horizons

Final-time fairness

Require fairness only for A^T .

Prefix-wise fairness

Require fairness for every A^t , $t \leq T$.

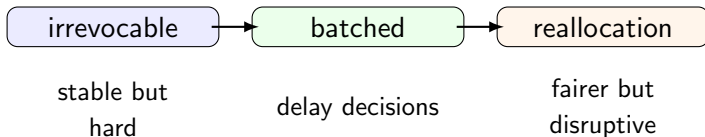
Asymptotic fairness

Require normalized unfairness to vanish, e.g.

$$\text{Envy}_T / T \rightarrow 0.$$

Commitment Models

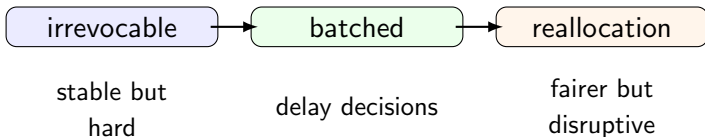
**How strongly an online algorithm is bound by its earlier allocation decisions.*



- irrevocable: $A_i^{t-1} \subseteq A_i^t, \forall i \in N$; easy to implement; no reassignment cost.
- batched: **waits** until a collection of goods has arrived.
- reallocation: an algorithm is allowed to **repair** earlier decisions; additional cost.

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A major question: how much “changing the past” is necessary for fairness?

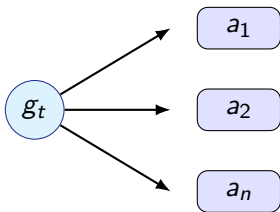
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Item-Centric Mechanisms

Offline algorithms often ask: *which good does this agent pick?*

Online algorithms often ask: *which agent should receive this arriving good?*

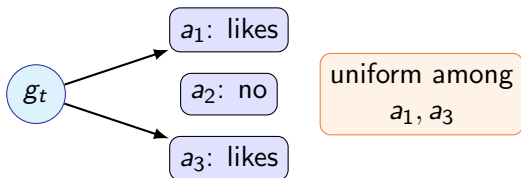


choose eligible agents
then maybe randomize

LIKE Mechanism ([Aleksandrov et al. 2015] & [Aleksandrov & Walsh 2020])

Definition (LIKE)

When g_t arrives, allocate it uniformly at random among agents who report positive value for it.



- simple;
- online strategy-proof in one-step binary settings;
- envy-free ex ante in some restricted domains.

Balanced LIKE

Definition (BALANCED LIKE)

*Among agents who like the item, allocate uniformly to **those with the fewest** previously allocated liked items.*

Design goal

Balance the number of allocated liked items over time.

Strategic caveat

An agent may skip a current liked item to become more eligible for future items.

One-Step Strategy-Proofness of LIKE

Proposition

*In **binary** utilities, truthful reporting is weakly optimal for the current item under LIKE.*

Proof.

If $v_i(g_t) = 0$, receiving the item gives utility 0, so reporting yes or no gives current utility 0.

If $v_i(g_t) = 1$, reporting no gives probability 0 of receiving g_t . Reporting yes gives probability $1/k$ for some $k \geq 1$, hence expected utility $1/k > 0$. $\square$

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Cumulative Envy

Definition (Envy after T rounds)

$$E_{ij}^T = v_i(A_j^T) - v_i(A_i^T),$$
$$\text{Envy}_T = \max_{i,j} \max\{0, E_{ij}^T\}.$$

Vanishing envy [Benade et al. (EC'18)]

An algorithm has vanishing envy if

$$\text{Envy}_T / T \rightarrow 0.$$

Random Allocation: A Clean Bound

Theorem

If $v_i(g_t) \in [0, 1]$ and each item is allocated uniformly at random, then with probability at least $1 - \delta$,

$$\text{Envy}_T \leq \sqrt{2T \log \frac{n(n-1)}{\delta}}.$$

Consequence

$$\frac{\text{Envy}_T}{T} = O\left(\sqrt{\frac{\log n}{T}}\right)$$

with high probability.

Proof of the Random Allocation Bound

For a fixed pair (i, j) , define

$$X_t = v_i(g_t)(\mathbf{1}\{g_t \rightarrow j\} - \mathbf{1}\{g_t \rightarrow i\}).$$

Then:

- $E_{ij}^T = \sum_{t=1}^T X_t$;
- $\mathbb{E}[X_t] = 0$;
- $X_t \in [-1, 1]$.

Hoeffding gives

$$\Pr[E_{ij}^T \geq r] \leq e^{-r^2/(2T)}.$$

Union bound over $n(n-1)$ ordered pairs and set

$$r = \sqrt{2T \log \frac{n(n-1)}{\delta}}.$$

Optimal Vanishing Envy

Theorem (Benade et al. (EC'18))

For adversarial online allocation of T goods with values in $[0, 1]$, there is a deterministic polynomial-time algorithm with

$$\text{Envy}_T \in \tilde{O}(\sqrt{T/n}),$$

and this is asymptotically optimal up to logarithmic factors.

Methodological lesson

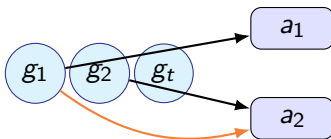
Exact fairness may be impossible online, but normalized unfairness can vanish.

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Changing the Past

- Requirement: after every time t , allocation A^t is EF1.
- To make this possible, old goods may be reassigned.
- **Objective: minimize the number of adjustments.**



moving g_1
is an adjustment

EF1 with Adjustments: Main Theorem

Theorem (He–Procaccia–Psomas–Zeng @IJCAI'19)

For online additive goods with EF1 required after every round:

- ① *two agents: informed algorithms need no adjustments, but uninformed algorithms need $\Theta(T)$;*
- ② *three or more agents: even informed algorithms need $\Omega(T)$ adjustments;*
- ③ *for $n \geq 3$, an uninformed algorithm uses $O(T^{3/2})$ adjustments.*

Interpretation of the EF1 Adjustment Result

- With two agents, future information can eliminate instability.
- Without future information, the adversary can force repeated repairs.
- With ≥ 3 agents, pairwise envy constraints interact cyclically.
- The online objective becomes bicriteria:
fairness quality + stability of past allocations.

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Online MMS Benchmark

Offline MMS is defined using all goods:

$$\mu_i^n(M) = \max_{(B_1, B_2, \dots, B_n)} \min_k v_i(B_k).$$

In online allocation, the algorithm does not know final M_T .

Definition (α -competitive online MMS (Zhou–Bai–Wu @ICML'23))

An online algorithm is α -competitive for MMS goods if

$$v_i(A_i^T) \geq \alpha \mu_i^n(M_T) \quad \forall i.$$

Online MMS for Goods

Theorem (Zhou–Bai–Wu @ICML'23)

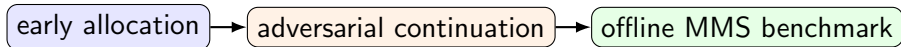
For online additive indivisible goods:

- *no positive competitive ratio is possible for MMS when $n \geq 3$;*
- *for two agents, there is an optimal $1/2$ -competitive algorithm.*

Takeaway

MMS is a global benchmark. Adversarial online arrivals can make it unattainable for ≥ 3 agents.

Why MMS Is Fragile Online



the future can make an early losing agent's MMS positive

Irrevocable decisions cannot be corrected after the final benchmark becomes clear.

Goods vs. Chores

Theorem (Zhou–Bai–Wu @ICML'23)

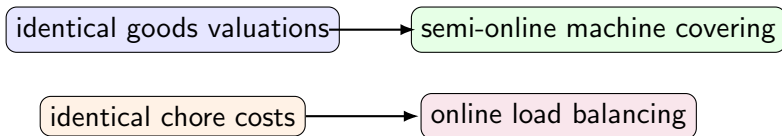
For online chores:

- *there is a $(2 - 1/n)$ -competitive algorithm for $n \geq 3$;*
- *there is a $\sqrt{2}$ -competitive algorithm for two agents;*
- *no algorithm can do better than $15/11$ for two agents.*

Reason for different behavior

For chores, load-balancing analogies provide stronger online structure.

Scheduling Connection



Online fair division inherits tools and lower bounds from online scheduling.

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Binary Valuations

Definition (Binary valuations)

$$v_i(g) \in \{0, 1\}.$$

- A good matters only to agents who like it.
- LIKE-type eligibility rules become natural.
- Fairness can often be expressed in terms of balancing liked items.

	g_t	eligible?	mechanism
a_1	1	yes	maybe receives
a_2	0	no	excluded
a_3	1	yes	maybe receives

Binary and Bivalued Online Allocation

Theorem (Wang–Wei @AAAI'26)

For online goods and chores with binary or personalized bivalued additive values, several EF1/MMS/welfare combinations become achievable, and counterexamples show the limits of these guarantees.

Why restrictions help

When each item has only a few possible values, local balancing can control global fairness more effectively.

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Honor Among Bandits: Why Relevant?

Problem type

Online fair allocation of indivisible goods with unknown stochastic values.

- There are n players and m item types.
- At each round t , an item of type $k_t \in [m]$ arrives.
- The item must be allocated immediately to one player.
- Player i 's value for type k has unknown mean μ_{ik}^* .
- The algorithm observes only the reward of the selected player.

Procaccia, Schiffer, & Zhang @NeurIPS 24

online fair division + unknown valuations + bandit feedback.

Model: Fractional Allocation Policy

A fractional allocation is a matrix

$$X \in \mathbb{R}_{\geq 0}^{n \times m}, \quad \sum_{i=1}^n X_{ik} = 1 \quad \forall k \in [m].$$

- X_{ik} is the probability that a type- k item is allocated to player i .
- If the true mean matrix is μ^* , the expected welfare is

$$\langle X, \mu^* \rangle = \sum_{i=1}^n \sum_{k=1}^m X_{ik} \mu_{ik}^*.$$

At round t :

k_t arrives, $i_t \sim X_{:,k_t}$, observe reward of (i_t, k_t) .

Fairness in Expectation

The paper studies welfare maximization subject to fairness constraints defined using the unknown mean matrix μ^* .

Envy-freeness in expectation

For every pair of players i, j ,

$$\sum_{k=1}^m X_{ik} \mu_{ik}^* \geq \sum_{k=1}^m X_{jk} \mu_{ik}^*.$$

Player i does not prefer player j 's allocation in expectation.

Proportionality in expectation

For every player i ,

$$\sum_{k=1}^m X_{ik} \mu_{ik}^* \geq \frac{1}{n} \sum_{k=1}^m \mu_{ik}^*.$$

Regret Benchmark

If μ^* were known, one could solve the fair welfare maximization LP:

$$Y_{\mu^*} \in \arg \max_{X \in \mathcal{F}(\mu^*)} \langle X, \mu^* \rangle,$$

$\mathcal{F}(\mu^*)$: the set of feasible fractional allocations satisfying the fairness constraints.

Regret

For an online algorithm choosing $X_1, \dots, X_T$,

$$R(T) = T \langle Y_{\mu^*}, \mu^* \rangle - \sum_{t=1}^T \mathbb{E}[\langle X_t, \mu^* \rangle].$$

Goal: low regret while satisfying fairness constraints.

Main Result: Explore-Then-Commit

Algorithmic idea

- 1 **Explore:** allocate items to estimate the unknown means μ_{ik}^* .
- 2 **Construct confidence intervals:** use the estimates to build a safe approximation of the fair feasible region.
- 3 **Commit:** solve an optimistic fair LP and use its allocation policy for the remaining rounds.

Guarantee

For envy-freeness in expectation or proportionality in expectation, the explore-then-commit algorithm achieves $\tilde{O}(T^{2/3})$ regret while maintaining the fairness constraint.

The paper also proves a matching $\tilde{\Omega}(T^{2/3})$ lower bound for this setting.

Bandit Learning Result

Theorem (Schiffer–Zhang @AAAI'2025)

There is a UCB-style online fair division algorithm that satisfies proportionality constraints with high probability and achieves

$$\tilde{O}(\sqrt{T})$$

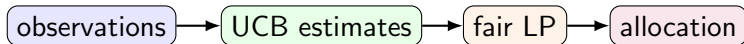
welfare regret.

Key improvement

This improves the previous best regret rate of $\tilde{O}(T^{2/3})$.

Why UCB Can Work with Fairness Constraints

- UCB estimates unknown values optimistically.
- Linear optimization enforces proportionality constraints.
- Confidence intervals keep constraint violations controlled.
- The main challenge is many potentially tight fairness constraints.



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Stochastic Online Fair Allocation

- Items may be drawn from distributions.
- Values may be known distributions or unknown means.
- The algorithm can target expected fairness and welfare regret.

Typical goal

Maximize welfare while satisfying **proportionality** or **envy** constraints **in expectation or with high probability**.

Bandit Learning Model

- Finite item types.
- Values are drawn from distributions with unknown means.
- The algorithm learns while allocating.

Definition (Regret)

Regret compares the algorithm's welfare to the best fair allocation policy with known means.

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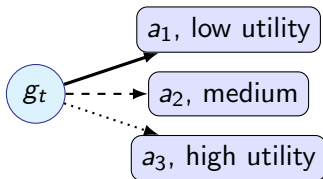
Nash Welfare in Online Allocation

Definition (Nash welfare)

$$\text{NSW}(A) = \left(\prod_{i=1}^n v_i(A_i) \right)^{1/n}.$$

- Offline MNW is EF1 and Pareto optimal.
- Online algorithms use accumulated utilities to guide allocation.
- Long-run guarantees are often asymptotic rather than exact.

PACE: Pacing According to Current Estimated Utility



lower accumulated utility
gets larger pacing weight

Best-of-Many-Worlds Guarantee

Theorem (Yang–Liao–Kroer 2024)

PACE/integral greedy allocation achieves asymptotically bounded multiplicative envy and competitive Nash-welfare guarantees under adversarial inputs with natural restrictions, complementing stochastic guarantees.

Best-of-many-worlds

One simple dynamic performs well under stationary, nonstationary stochastic, and adversarial input models.

Multiplicative Envy

Definition (Multiplicative envy)

A typical multiplicative envy ratio compares

$$\frac{v_i(A_j)}{v_i(A_i)}.$$

- Additive envy asks for differences.
- Multiplicative envy asks for ratios.
- Ratio notions interact naturally with Nash welfare.

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Online Fair Division with Subsidies

Online model

Indivisible goods $g_1, g_2, \dots$ arrive sequentially. When g_t arrives:

- its value information becomes available;
- it must be allocated immediately and irrevocably;
- the number of future goods is unknown.

Let $X^t = (X_1^t, X_2^t, \dots, X_n^t)$ denote the allocation after the first t goods.

Definition (Envy-freeness with subsidy)

Given a nonnegative subsidy vector $p^t = (p_1^t, p_2^t, \dots, p_n^t) \in \mathbb{R}_{\geq 0}^n$, the outcome (X^t, p^t) is envy-free if

$$v_i(X_i^t) + p_i^t \geq v_i(X_j^t) + p_j^t \quad \forall i, j \in N.$$

An allocation X^t is **envy-freeable** if such a subsidy vector exists. [Kulkarni, Mehta, Narayan, & Ponitka, @AAMAS 2026.]

Envy-Freeability and Local Efficiency (1/2)

Definition (Local efficiency)

An allocation $X = (X_1, X_2, \dots, X_n)$ is locally efficient if no permutation of its bundles increases social welfare:

$$\sum_{i \in N} v_i(X_i) \geq \sum_{i \in N} v_{\pi(i)}(X_i) \quad \text{for every permutation } \pi : N \rightarrow N.$$

Theorem (Envy-freeability characterization)

An allocation is envy-freeable if and only if it is locally efficient.

Online implication

Because the arrival process may terminate after any round, an online algorithm must maintain local efficiency after every prefix:

X^t must be locally efficient for every t .

Envy-Freeability and Local Efficiency (2/2)

Theorem (Additive valuations (Kulkarni et al., @AAMAS 2026))

For additive valuations, when good g_t arrives, assigning it to

$$i^* \in \arg \max_{i \in N} v_i(g_t)$$

preserves local efficiency and therefore preserves envy-freeability.

When Can Envy-Freeability Be Maintained Online?

Valuation class	Online envy-freeability
Additive	Yes
SPLC	Yes
k -demand	Yes
Budget-additive	No in general
Submodular with binary marginals	No in general
Supermodular with binary marginals	No in general

Subsidy cost may be large

Assuming every marginal value is at most 1, the total subsidy needed online can be as large as $\Omega(m(n-1))$ even for additive valuations.

Classes with item-independent subsidy bounds (Kulkarni et al., @AAMAS 2026))

Bounds independent of m are obtained for structured classes, including fixed- k demand, fixed- k valued, rank-one, restricted additive, and identical monotone valuations.

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Divisible Online Allocation

- Some online fair allocation work studies **divisible** resources.
- Divisibility avoids single-good impossibilities but retains online uncertainty.
- With normalized values, nontrivial fairness–efficiency tradeoffs become possible.

Example

Gkatzelis–Psomas–Tan (AAAI'21) study immediately and irrevocably allocated divisible resources under normalized valuations.

Learning-Augmented Fairness

- Prediction may concern future arrivals, item types, values, or the optimal outcome.
- Desired guarantees:

Consistency

Good performance when predictions are accurate.

Robustness

Worst-case protection when predictions are wrong.

Open direction: consistency–robustness for online indivisible goods.

Reference: Aziz, Guo, Lam, Zhou @NeurIPS'25.

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Open Problem: Online EF1

Open Problem

For which valuation domains and arrival models can an irrevocable online algorithm guarantee final-time or prefix-wise EF1?

- General additive valuations seem difficult.
- Binary and bivalued valuations are promising.
- Limited reallocation may give a useful middle ground.

Open Problem: Online MMS

Open Problem

After the no-competitive-ratio result for goods with $n \geq 3$, what relaxations make online MMS meaningful?

- stochastic arrivals;
- predictions of future types;
- resource augmentation;
- bounded reallocation;
- restricted valuations.

Open Problems: Online Fairness with Subsidies

Open Problem

For which valuation classes can an online algorithm maintain envy-freeability with total subsidy independent of the number of arriving goods?

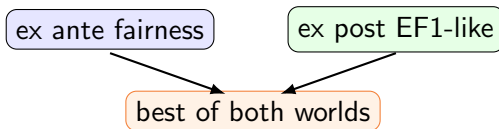
- Characterize the boundary between valuation classes that admit online envy-freeability and those that do not.
- Determine tight subsidy bounds for broader valuation classes.
- Study the trade-off between subsidy, welfare, and online fairness.
- Consider models in which the subsidy budget is fixed in advance.
- Study truthful online mechanisms when allocation and subsidy decisions are both strategic.

[Motivated by Kulkarni et al., @AAMAS 2026.]

Open Problem: Best of Both Worlds

Open Problem

Can we obtain ex ante fairness and ex post approximate fairness simultaneously in online indivisible allocation?



Summary

- Online fair allocation differs fundamentally from offline fair division.
- EF1 interacts naturally with reallocation and stability costs.
- MMS is fragile under adversarial irrevocable arrivals.
- Subsidies can restore exact envy-freeness online, but envy-freeability and subsidy cost depend sharply on the valuation structure.
- Stochastic, bandit, and pacing models recover strong long-run guarantees.
- Learning-augmented online fair allocation remains underdeveloped.

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Discussions